Cheat sheet: Week 5 Wednesday, what the model is allowed to know



The question: on the morning the model is used, do we know this value for this customer?

Term	Plain meaning	Rule or formula	Trap
Recency	days since the last order	`prediction morning - last order`	computed from today, it leaks
Ratio feature	puts old and new customers on one scale	`orders / months as a member`	a total ranks the oldest first
Cohort age	how long a customer has been with us	`prediction morning - first order`	an id issued in order hides it
One-hot	one 0 or 1 column per category	for linear and distance models	a 1, 2, 3 code invents an order
Standardise	put numbers on one scale	`z = (value - mean) / std`, fitted on training rows	fitting on all rows lets test rows in
Sentinel	a code meaning "never", such as 999	split into a flag plus the number	its mean describes nobody
Leakage	an input unknown at prediction time	ask the prediction-time question	"too good" starts the check, never ends it
Calendar leak	a column that stamps the period	test with a split by time	known that morning and still borrowed
As-of trap	today's value of an overwritten field	read a snapshot from the morning	"it is in the warehouse"
Permutation importance	score lost when one column is shuffled	on validation rows; twins as a group	ranks reliance, never cause
Sign flip	a coefficient changes sign when a partner joins	read correlated columns as a group	"the new sign is the true effect"

When	Do this
A score jumps on one new column	find its write time before anything else
Logistic or distance model	one-hot and standardise inside the pipeline
Tree model	skip scaling; an ordinal code is fine
A field is overwritten nightly	ask for history and read it as of the morning
One importance bar towers	check for a leak, then for close partners

The bank's numbers to carry: AUC 0.761, 0.795, 0.932 with `duration`; honest 0.796, about 0.57 inside one month; bill Rs 1.19 lakh promised against Rs 2.82 lakh real; euribor3m -0.92 alone, +0.77 beside its partners.

Everything so far, one line each

Day	The line
W1 Mon	a number is a tree of drivers; refuse the average when one customer buys in bulk
W1 Tue	compare like with like, with the denominator named
W1 Wed	profile before you analyse; input equals clean plus rejected
W1 Thu	chance first: the permutation test, and correlation is not cause
W2 Mon to Wed	SQL from selection to window functions; joins that do not fan out
W2 Thu	groupby, merge and reshape build the customer table
W2 Fri	one number reaches the deck, recalculating live
W4 Mon	a metric with guardrails, so it cannot be gamed
W4 Tue	one slide, five minutes, three people paid to disagree
W4 Wed	lift: pairs beyond chance, valued in rupees
W4 Thu	cohorts and funnels: where the loss happens
W4 Fri	forecasts against a baseline that is hard to beat
W5 Mon	frame the target and the moment; beat the 0.887 baseline
W5 Tue	recall and the bill; threshold 40 / 1,240 = 0.032; precision at k under a budget
W5 Wed	every feature with a reason, every column with a write time, importance as reliance

Three things that happened

Story	The lesson	Source, verified 28 October 2026
A pneumonia network learned which hospital took an X-ray, partly from a metal token: AUC 0.931 at home, 0.815 at a new site	a column known in time can still stand in for a base rate	https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.1002683 (verified 28 October 2026)
Kaggle's 2011 social network winners re-identified the test set from a public crawl: AUC 0.981	ask where every input comes from at prediction time	https://arxiv.org/abs/1102.4374 (verified 28 October 2026)
Airbnb's histograms showed monthly prices logged as nightly ones for long stays	plot every feature before you trust it	https://arxiv.org/abs/1810.09591 (verified 28 October 2026)